Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 18 – Digital intimacy I



Content warning: repeated mentions of sexual abuse and violence

Last time

- New threat model: locally powerful adversaries
- Discussion on at-risk users
  - How to define at-risk users
  - Idea of at-risk framework as a living document, rather than a prescriptive and stagnant tool
    - How would the categories change when we look at emerging technologies?
  - Intersectionality
    - Are there certain combinations of factors that would increase risk?

Today

- Introduction to digital intimacy
  - Overview of the area and its importance in the field
  - Online dating
    - Core research questions that people study
  - Sex work
    - History of sex work research in digital safety leading to work on AI-enable sexual abuse
    - Core research questions that people study
  - Lessons across multiple groups where technology facilitates primarily offline interactions/relationships.

Digital intimacy research

Digital intimacy is the study of how technology mediates close personal connection (romantic, sexual, emotional, and relational) and the safety, security, and privacy risks that arise when it does.

Areas of focus include:

- Online dating
- Digitally-mediated consent
- Sex work
- AI-enabled sexual abuse
- AI in intimate relationships

Harms studied in digital intimacy are not new

- Danger from meeting strangers
- Blackmail and extortion
- Nonconsensual sharing of intimate photos
- Deception about identity (catfishing)
- Sexual violence
- Stalking and harassment
- …

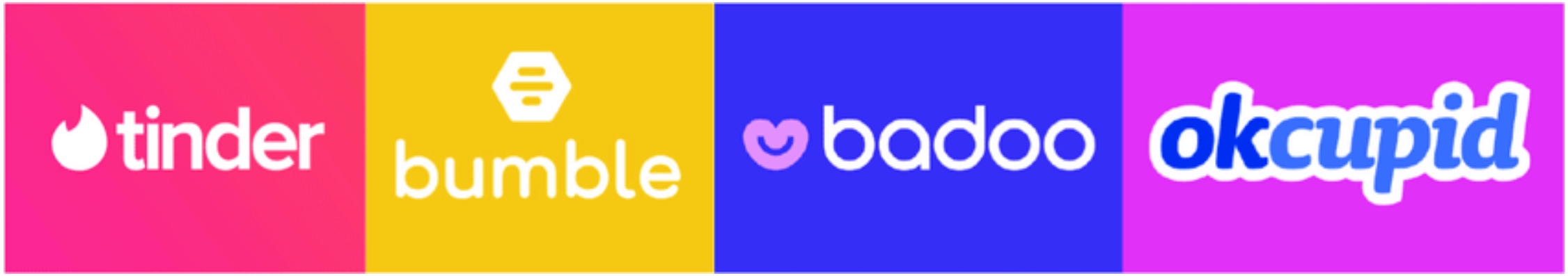
Digital intimacy research goals: (1) to understand how technology changes existing intimate harms, (2) to explain why those persist, and (3) to intervene

Importance of digital intimacy

- Technology is a huge part of people's personal and intimate lives
  - Meeting online is now the most common way U.S. couples meet & meeting through friends has been declining since 1995 (Rosenfeld et al., PNAS 2019)
- People are engaging with emerging AI technologies in intimate ways
- Digital intimacy is just one way that people use technology to mediate in-person interactions → generalize results to other contexts
- Lots of current policy discussion, especially around AI-enabled sexual abuse

Location-based dating apps

Location-based dating apps connect users with people who are physically nearby, sometimes sharing real-time coordinates, crossed paths, or their search radius.



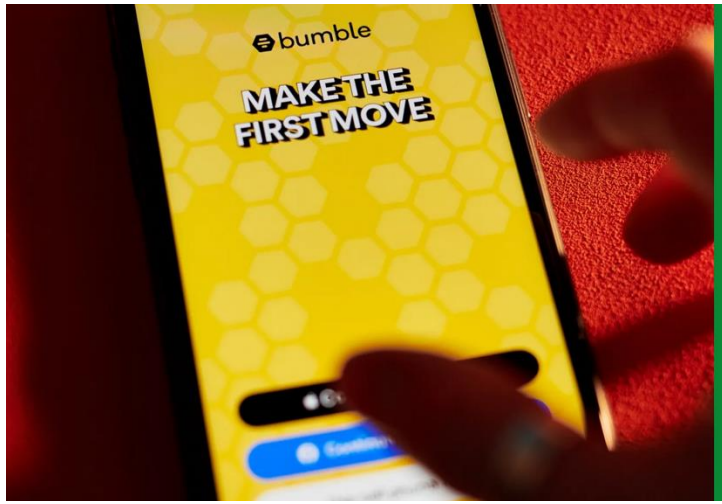
The New York Times

Queer Dating Apps Are Unsafe by Design

Privacy is particularly important for L.G.B.T.Q. people.

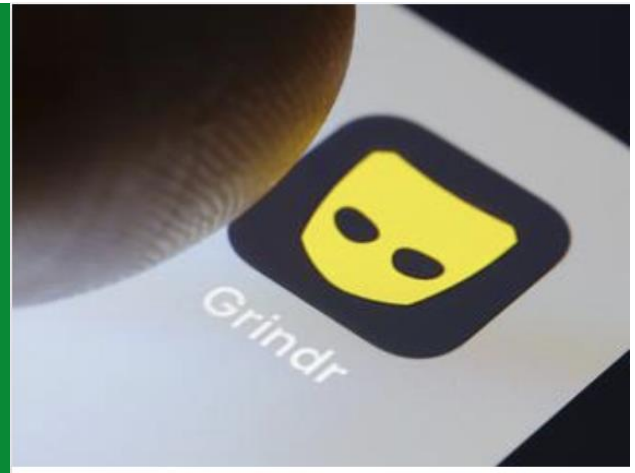
People ENTERTAINMENT CRIME HUMAN INTEREST LIFESTYLE ROYALS STYLEWATCH SHOPPING SUBSCRIBE

Woman Accused of 'Romance Scam on Steroids' After Allegedly Drugging Older Men in Deadly Dating App Scheme



SECURITY

Bumble and Hinge allowed stalkers to pinpoint users' locations down to 2 meters, researchers say



Lifestyle

ANALYSIS

Dating apps 'can damage mental health and body image'

Millions of people globally use dating apps - but what impact do they have on users? **Zac Bowman** reports

Hate crime unit investigates assaults linked to dating apps

Assaults linked to online dating meet-ups have been reported across the northern beaches and western Sydney, with the NSW Police hate crime unit stepping in and warnings issued to those using online apps.

Prior work on online dating has studied:

Platform-level privacy

What kinds of privacy risks should we be concerned about in online dating apps?

Experiences of LGBTQ+ users disclosing sensitive attributes

How do LGBTQ+ daters' risks and protective strategies differ from other daters'?

Gendered harassment on dating apps

What does staying safe cost users, especially women? (e.g., safety labor)

Consent and design interventions

Can dating apps be designed to prevent harm, especially sexual violence? How should dating apps mediate consent & risk assessment?

AI in dating

How are people using AI to make decisions in their romantic relationships? What is the impact of this on the relationship?

History of sex work research in digital safety

- Sociologists and criminologists studied how the internet changed sex work (pre-2017)
  - Takeaway: online advertising gave sex workers more control over their work, but technology also enabled new kinds of abuse
- HCI: Sex work through a social justice lens (2017-2019ish)
  - Coincided with HCI moving toward feminist and social-justice research
  - Takeaway: Safety is shaped by stigma, law, and power, not just technical design
- S&P: Sex work threat models & protective behaviors (2021-2024ish)
  - Establishing sex workers as an at-risk group with clear threat models: clients, weak legal protections, hostile platforms (McDonald et al., 2021, you read for today)
  - Shift to platform sex work (e.g., threat models of OnlyFans creators)

Prior work on sex work in digital safety:

Safety goals, threats, and protective practices

How do sex workers define safety, who they see as threats, and how they protect themselves?

Community-safety infrastructure

How do sex workers build their own safety tools (e.g., bad date lists and digital whisper networks), often in place of formal justice systems?

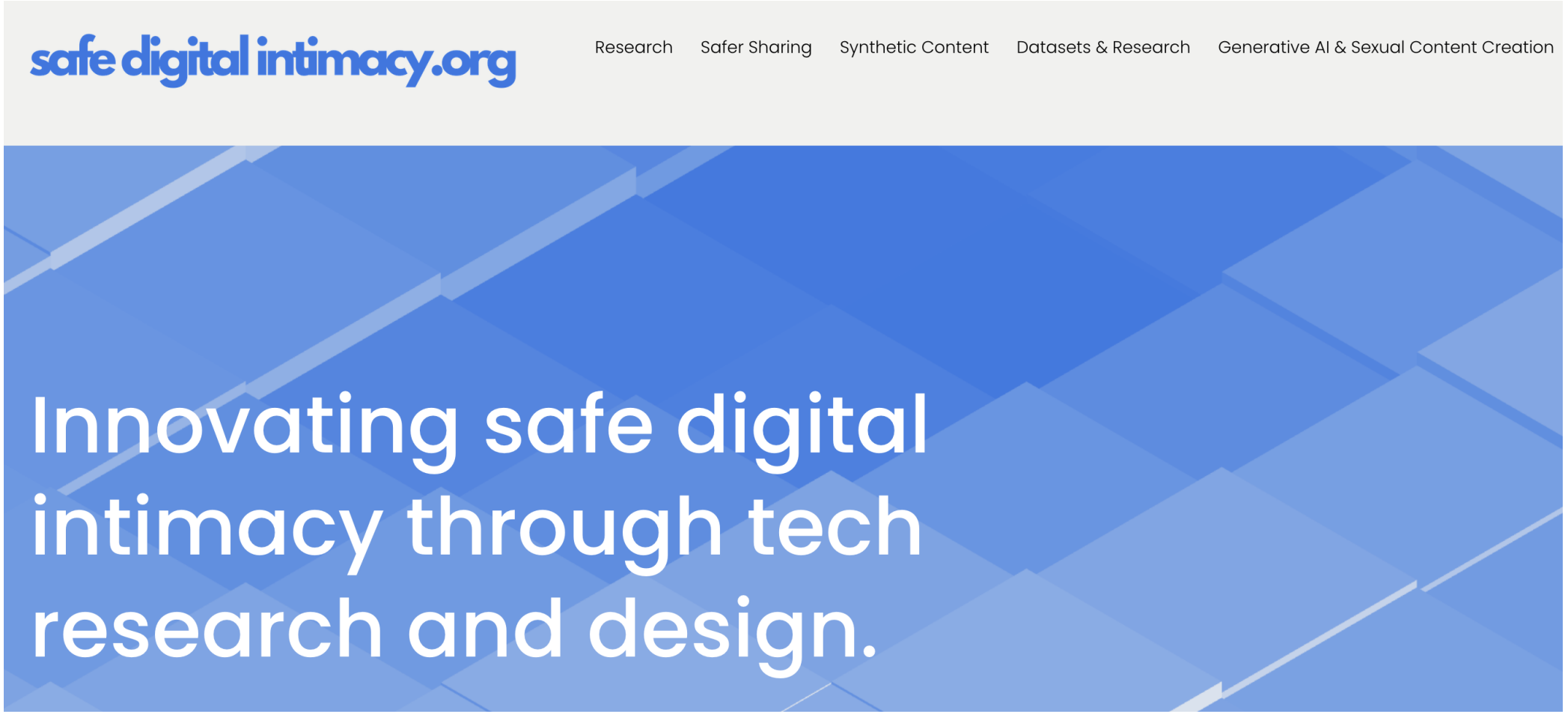
Platform governance and deplatforming

Impacts of platform policies, content moderation, and payment restrictions on sex workers

Digital and platform-mediated sex work

How does sex work intersect with online content creation? (e.g., content theft, stalking, doxxing, pressures to be publicly visible)

Impact of sex work research in HCI/S&P



Abstracting across groups



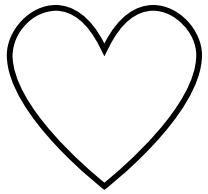
Algorithm matches strangers online

Strangers interact in the physical world

Algorithmically-mediated offline introduction (AMOI):

an **offline** introduction between strangers that is mediated by an **online** matching algorithm on a *digital platform*

We study this abstraction across three groups



Online daters



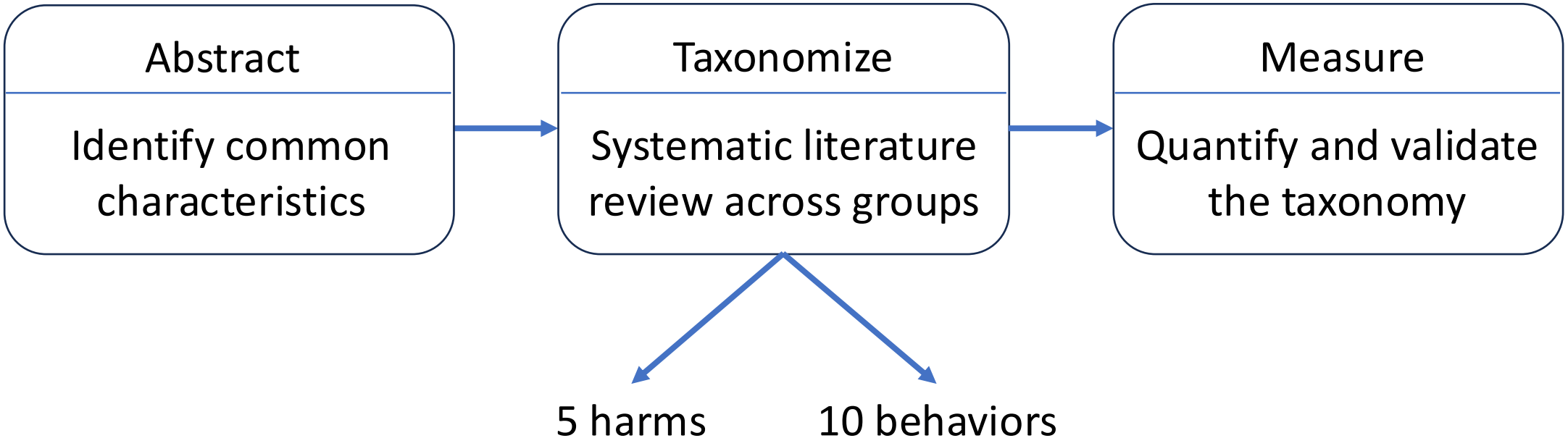
Sex workers



Gig workers

Veronica A. Rivera, Daricia Wilkinson, Aurelia Augusta, Sophie Li, Elissa M. Redmiles, Angelika Strohmayer. *Safer Algorithmically-Mediated Offline Introductions: Harms and Protective Behaviors*. Published at **CSCW 2024**.

Our approach: systematization + measurement



Measure: quantify and validate taxonomy



Survey

- 476 online daters
- 451 gig workers



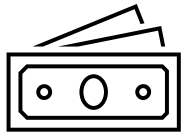
Measures

- Definitions of safety
- Experiences with harm
- Protective behaviors

Five harms



Physical



Financial



Privacy



Autonomy

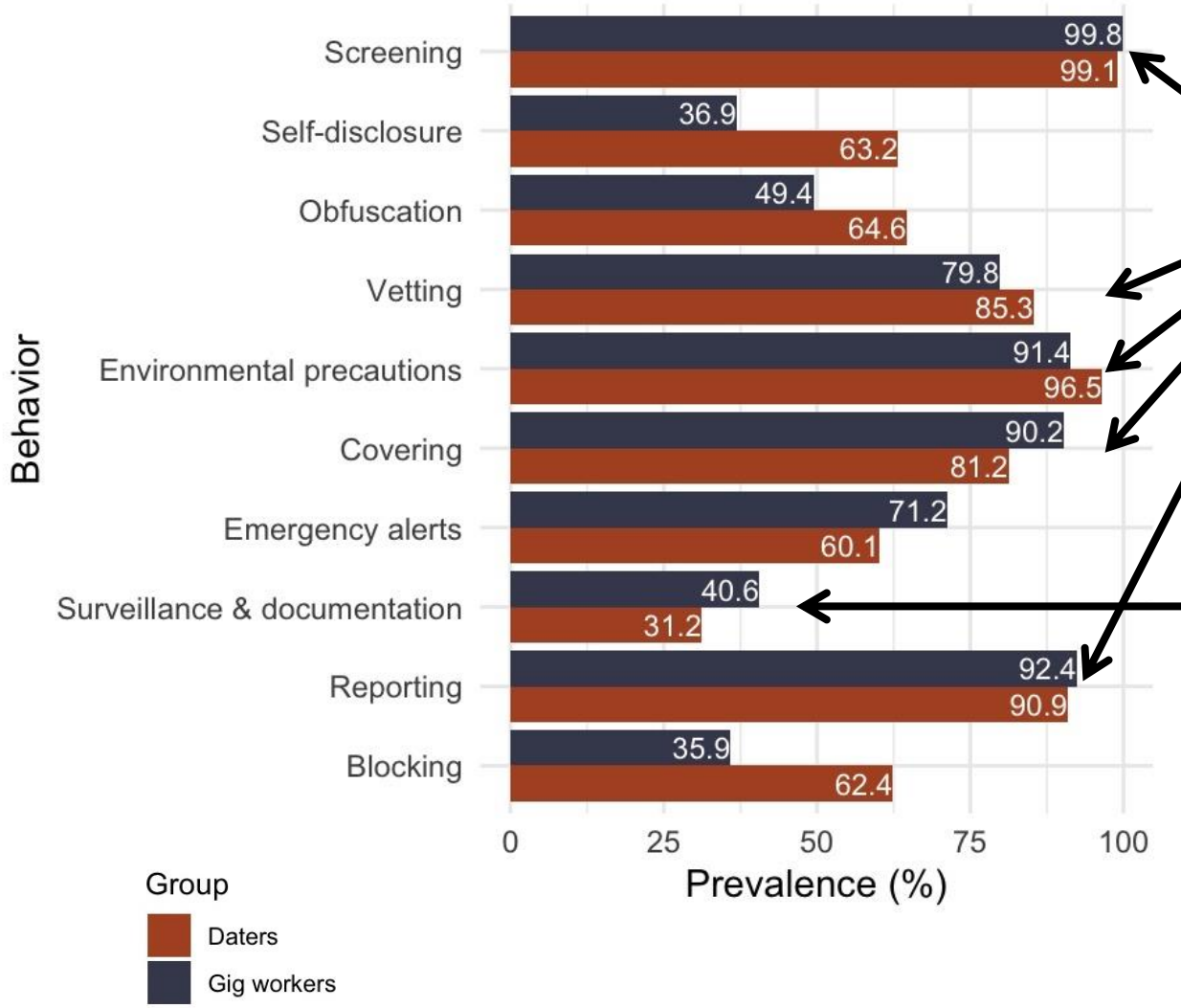


Emotional

Loss of control over private information

Loss of control over decision-making or physical body

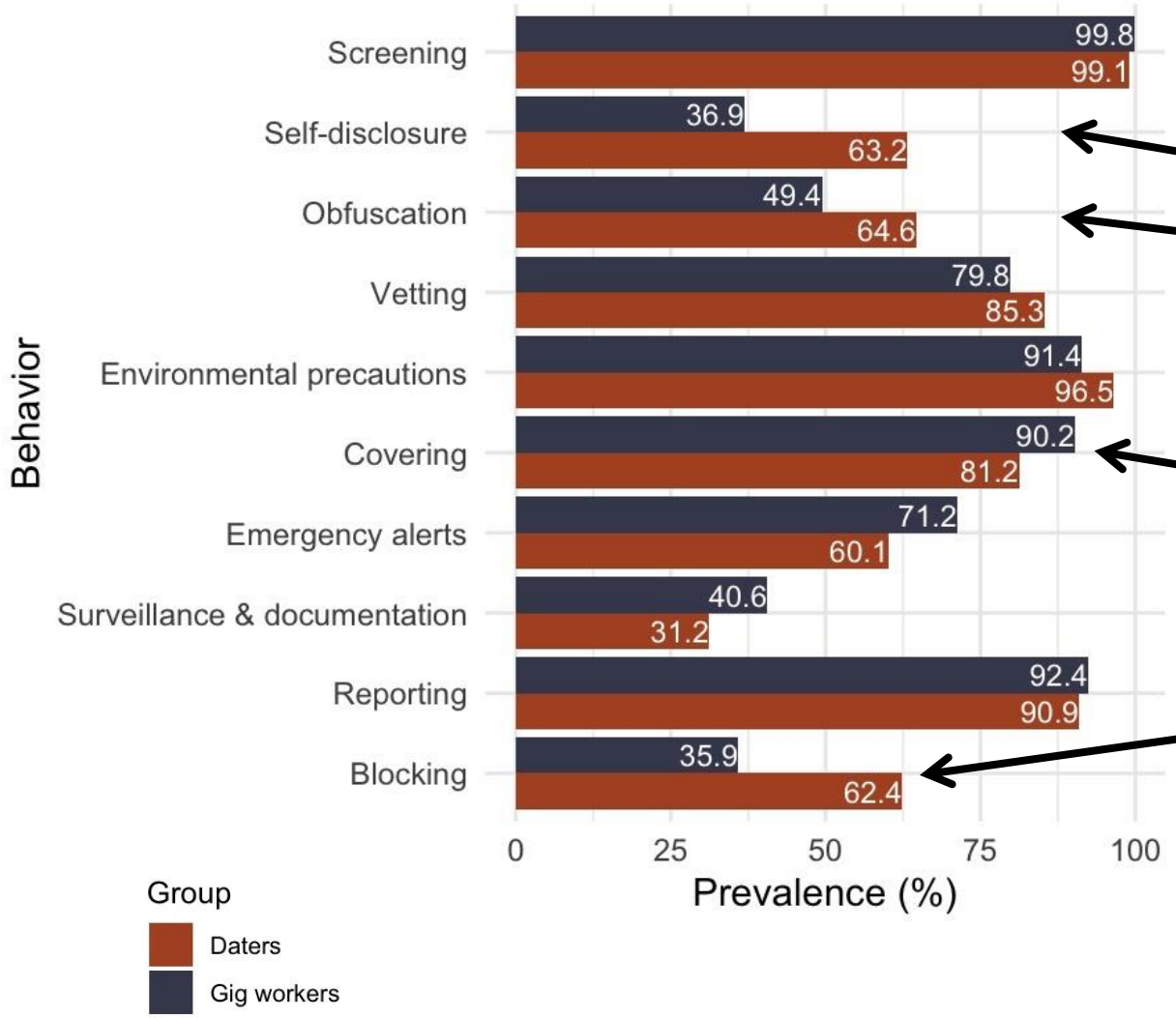
Users self-protection is ***pervasive***



Many behaviors are used by almost all users

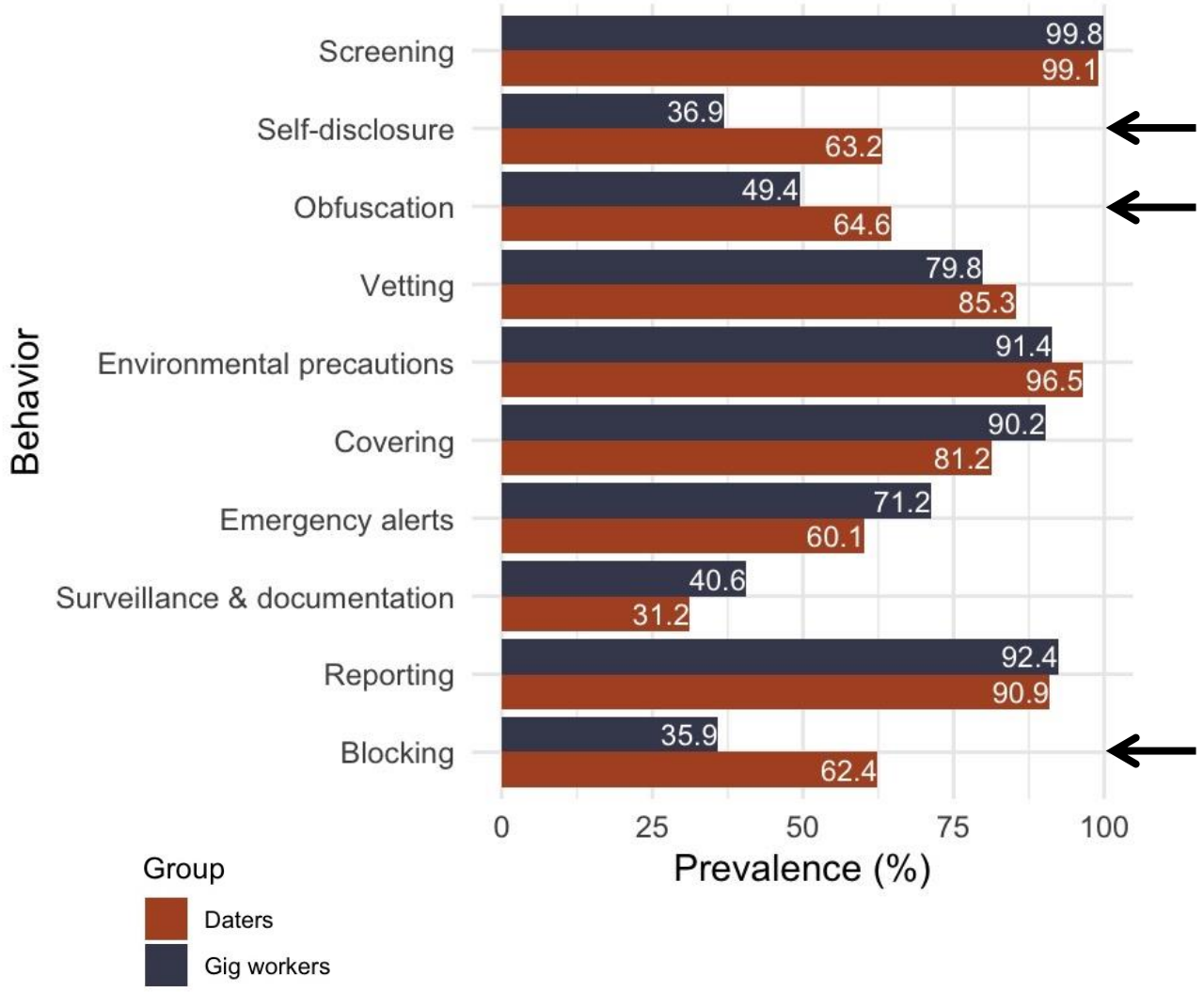
Every behavior is common

Differences are due platform design choices



Behavior	p-value
Screening	1.00
Self-disclosure	**< 0.01**
Obfuscation	**< 0.01**
Vetting	1.00
Environmental precautions	0.08
Covering	**0.03**
Emergency alerts	0.07
Documentation	0.10
Blocking	**< 0.01**
Reporting	1.00

Group differences suggest design directions



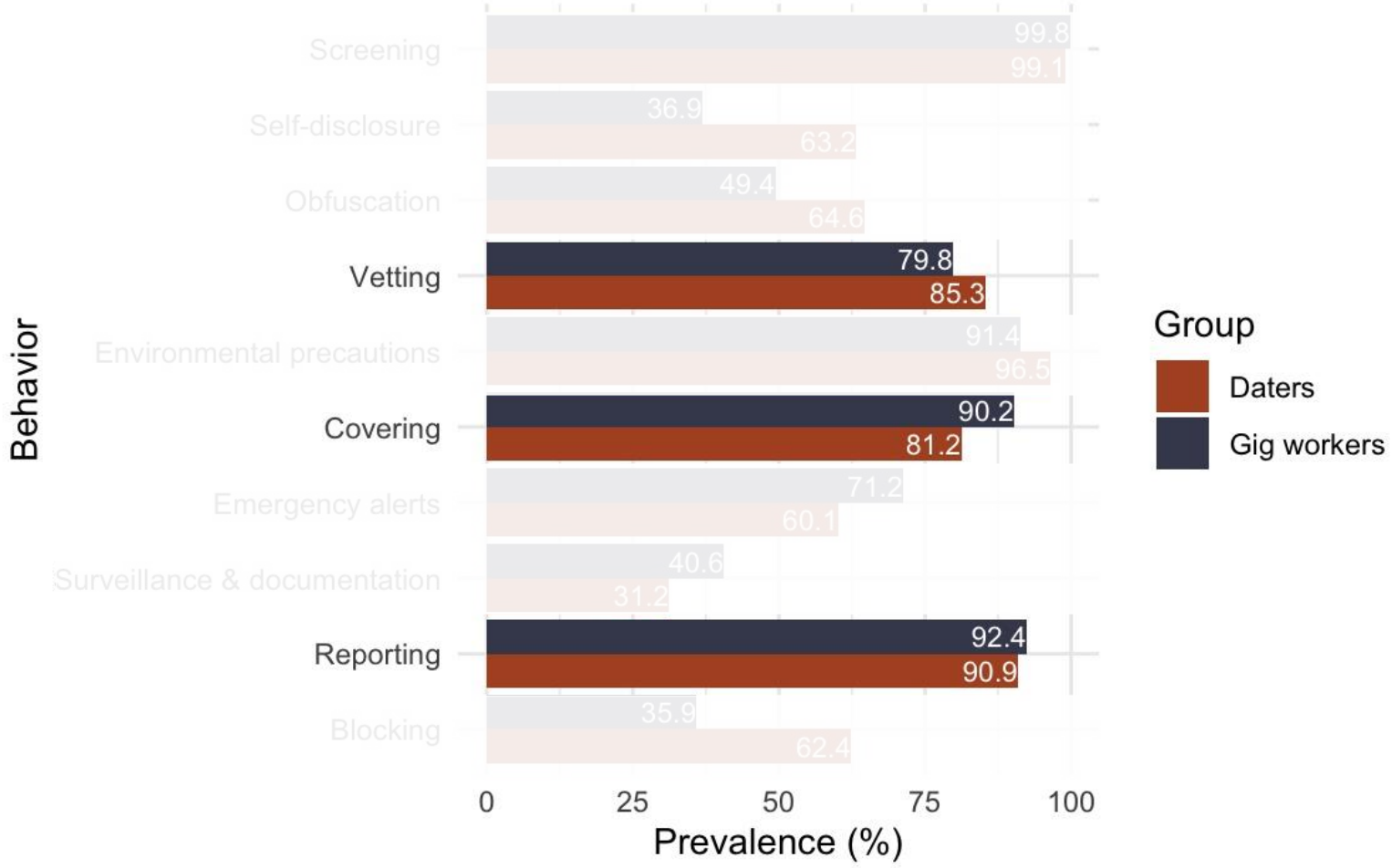
Blocking, self-disclosure, and obfuscation have the biggest **effect size**

Observation: gig platforms don't support these

Group similarities suggest design directions too

≥4/5 respondents use **social** protections:

- **vetting**
- **covering**
- **reporting**



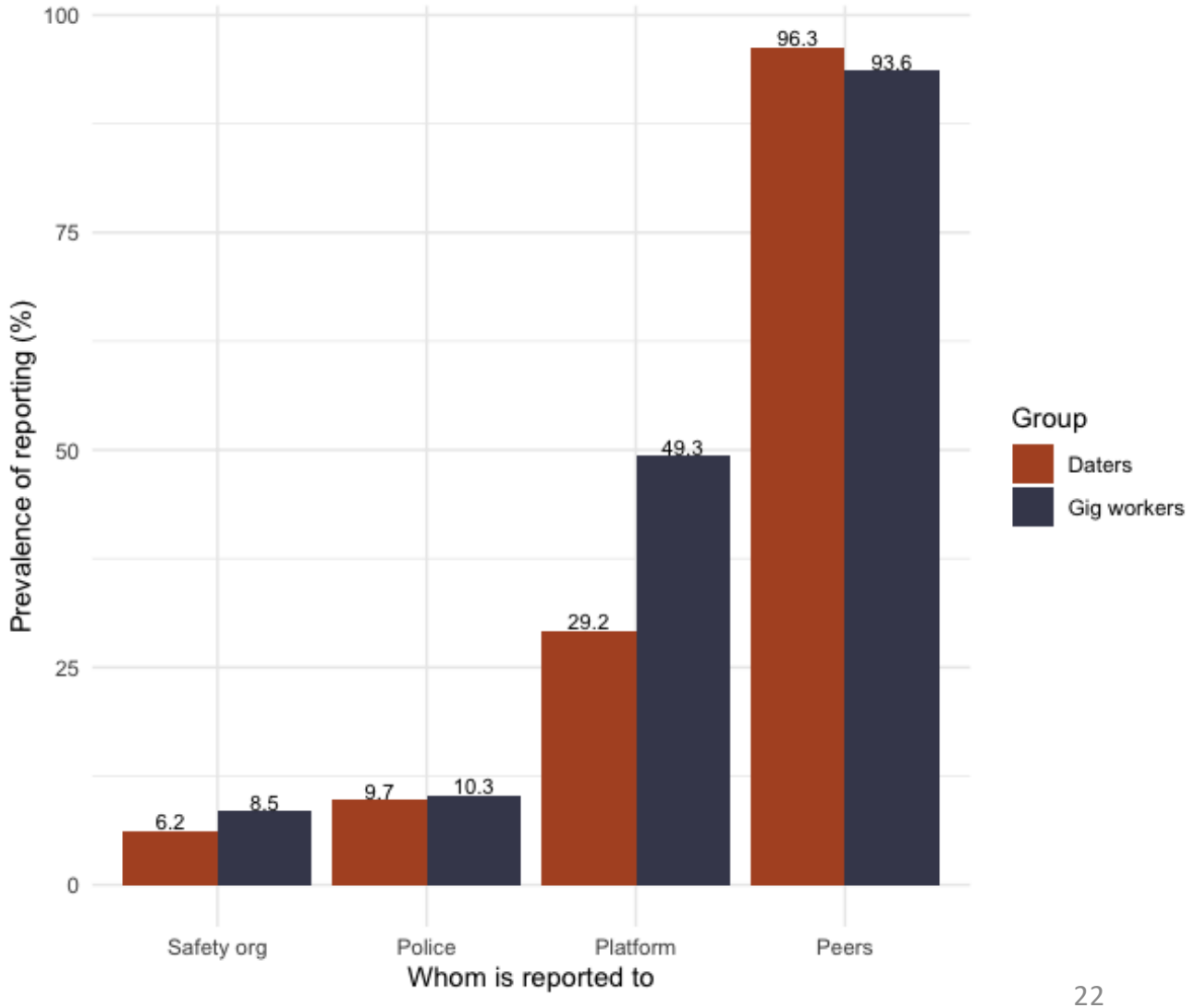
In practice, reporting harm is a **social** behavior

Many report to:

- Peers (>90%)

Few users report to:

- Platforms (< 30%; 50%)
- Police (< 10%)
- Safety NGOs (< 10%)



Announcements

- I graded and returned midpoint reports with comments last week
- I also shared a spreadsheet on Ed to sign up for a final presentation date and time.

Discussion

Warm up: Rank the different types of data leaks from most to least harmful. Be ready to explain your reasoning and how you are defining the harm that drives your ranking.

Table 3: Overview of data leaks for personal data.

	Tinder	Badoo	POF	MeetMe	Tagged	Grindr	Tantan	Jaumo	LOVOO	happn	Bumble	Hinge	Hily	OkCupid	Meetic
First name	◇	◇	○	◇	⟐	–	–	◇	◇	◇	◇	◇	◇	◇	◇
Last name	–	–	–	○	–	–	–	–	–	–	–	–	–	–	–
Gender	○	⟐	○	◇	⟐	○	⟐	⟐	◇	⟐	◇	○	⟐	◇	⟐
Age	◇	◇	◇	◇	◇	○	◇	◇	◇	◇	◇	◇	◇	◇	◇
Date of birth	–	–	–	–	⟐	–	–	–	–	–	–	○	–	–	–
Education	○	○	◇	○	–	–	○	⊙	○	○	○	○	◇	○	○
Employment	○	○	◇	–	–	–	○	⊙	○	○	○	○	◇	○	○
Languages spoken	–	○	○	–	○	–	–	⊙	○	–	○	○	○	○	○
Nationality	–	–	–	–	–	–	–	–	–	–	–	–	–	–	○
Place of residence	○	–	◇	–	◇	–	○	◇	◇	⊙	○	◇	–	–	◇
Hometown	–	–	–	–	–	–	–	–	–	–	○	○	–	–	–
Relationship status	–	⟐	○	○	○	○	–	⊙	○	–	–	–	–	○	○
Marital status	–	–	○	○	○	○	–	⊙	–	–	–	–	–	○	○
Having children	○	⟐	○	○	–	–	–	⊙	○	○	○	○	◇	○	○
Having siblings	–	–	◇	–	–	–	–	–	–	–	–	–	–	–	–
Email address	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Phone number	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Other platforms	–	–	–	–	–	○	–	–	–	–	–	–	⟐	–	–
Photos	◇	◇	○	○	○	○	◇	◇	◇	◇	◇	◇	◇	◇	◇
Interests	○	○	○	○	○	–	○	⊙	–	○	○	○	○	○	○
Income	–	–	–	–	–	–	○	–	–	–	–	–	–	–	○

Table 4: Overview of data leaks for sensitive data.

	Tinder	Badoo	POF	MeetMe	Tagged	Grindr	Tantan	Jaumo	LOVOO	happn	Bumble	Hinge	Hily	OkCupid	Meetic
Racial or ethnic origin	–	–	○	–	○	○	–	–	–	–	○	○	○	○	○
Political opinions	–	–	–	–	–	–	–	–	–	–	○	○	○	○	–
Religious/philos. beliefs	–	–	◇	–	○	–	–	⊙	–	–	○	○	○	○	○
Health data															
Height	–	⟐	◇	○	–	○	○	⊙	○	○	○	◇	○	○	○
Weight	–	–	–	–	–	○	–	–	–	–	–	–	–	–	○
Figure	–	–	○	○	–	○	–	⊙	–	–	–	–	○	○	–
Fitness	○	–	–	–	–	–	–	⊙	–	○	○	–	○	–	○
Diet	○	–	–	–	–	–	–	⊙	–	○	○	–	◇	○	○
Eye color	–	⊙	◇	–	–	–	–	–	–	–	–	–	–	–	○
Hair color	–	⊙	◇	–	–	–	–	–	–	–	–	–	–	–	○
Smoking	○	⟐	◇	○	–	–	–	⊙	○	○	○	○	◇	○	○
Alcohol	○	⟐	○	○	–	–	–	⊙	–	–	○	○	◇	○	–
Recreational drugs	–	–	◇	–	–	–	–	–	–	–	○	○	–	○	–
(COVID) vaccination	○	–	–	○	–	○	–	–	–	–	○	○	–	–	–
HIV status	–	–	–	–	–	○	–	–	–	–	–	–	–	–	–
Sexual orientation	○	⟐	⟐	⟐	○	–	–	–	–	–	○	–	⟐	–	
Sex life	–	–	–	–	–	○	–	–	–	–	–	–	◇	–	

Table 5: Overview of data leaks for app usage data.

	Tinder	Badoo	POF	MeetMe	Tagged	Grindr	Tantan	Jaumo	LOVOO	happn	Bumble	Hinge	Hily	OkCupid	Meetic
Other was recently active	◇	◇	◇	◇	◇	◇	◇	◇	◇	◇	⊕	◇	◇	◇	◇
Last activity time	–	⊕	⟐	⟐	◇	◇	◇	⟐	⟐	⟐	⊕	⟐	–	⟐	◇
Account creation time	–	–	–	–	◇	–	⟐	–	–	–	–	⟐	–	⟐	⟐
Relationship type sought	⟐	○	◇	◇	○	○	–	○	◇	○	○	○	◇	◇	○
Wanting children	○	⟐	○	–	–	–	–	⊙	–	○	○	○	◇	○	○
Filters	–	⟐	◇	⟐	–	–	⟐	◇	–	–	⊕	–	–	⟐	◇
# profiles per API request															
Card stack	10	20	69	1	20	–	5	10	30	20	10	20	15	1	30
Grid	–	20	42	30	24	100	–	–	40	–	–	10	–	–	20
Permanent profile access	⟐	◇	◇	◇	◇	◇	⟐	⟐	◇	⟐	⟐	⟐	⟐	◇	◇
See profiles while paused	⟐	–	◇	–	◇	–	–	⟐	–	⟐	–	⟐	–	–	–
Other has liked you	–	⟐	⟐	⟐	⟐	–	⟐	⟐	–	⟐	⟐	⟐	⟐	⟐	⟐
Other has disliked you	–	⟐	–	–	⟐	–	–	–	–	–	⟐	–	–	⟐	⟐
Popularity score	–	⟐	–	–	–	–	⟐	–	–	–	–	–	–	–	–
Number of likes/dislikes	–	⊕	–	–	–	–	⟐	–	–	–	–	–	–	–	–

Incidence	UI Exposure	Traffic Leak	Exfiltration Leak
Always	◇	⟐	⊕
If set/shown	○	⊙	⊕
Never	–		